

HW1 Sample answer

The answer is just for reference!

Exercise 1.1

We determine the parity by checking the function definitions.

1. f even, g even:

- $(f + g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f + g)(x) \implies \text{Even}$
- $(f \cdot g)(-x) = f(-x)g(-x) = f(x)g(x) = (f \cdot g)(x) \implies \text{Even}$
- $(f \circ g)(-x) = f(g(-x)) = f(g(x)) = (f \circ g)(x) \implies \text{Even}$

2. f even, g odd:

- $(f + g)(-x) = f(-x) + g(-x) = f(x) - g(x) \implies \text{Neither}$
- $(f \cdot g)(-x) = f(-x)g(-x) = f(x)(-g(x)) = -f(x)g(x) \implies \text{Odd}$
- $(f \circ g)(-x) = f(g(-x)) = f(-g(x)) = f(g(x)) \implies \text{Even}$

3. f odd, g even:

- $(f + g)(-x) = f(-x) + g(-x) = -f(x) + g(x) \implies \text{Neither}$
- $(f \cdot g)(-x) = f(-x)g(-x) = (-f(x))g(x) = -f(x)g(x) \implies \text{Odd}$
- $(f \circ g)(-x) = f(g(-x)) = f(g(x)) = (f \circ g)(x) \implies \text{Even}$

4. f odd, g odd:

- $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f + g)(x) \implies \text{Odd}$
- $(f \cdot g)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) \implies \text{Even}$
- $(f \circ g)(-x) = f(g(-x)) = f(-g(x)) = -f(g(x)) \implies \text{Odd}$

Summary Table:

f	g	$f + g$	$f \cdot g$	$f \circ g$
even	even	even	even	even
even	odd	neither	odd	even
odd	even	neither	odd	even
odd	odd	odd	even	odd

Exercise 1.2

The composition $(f \circ g)(x)$ is computed as follows:

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) \\
 &= f(m_2x + b_2) \\
 &= m_1(m_2x + b_2) + b_1 \\
 &= (m_1m_2)x + (m_1b_2 + b_1)
 \end{aligned}$$

The result is a linear function of x , so $f \circ g$ is also an affine function. The slope of its graph is m_1m_2 .

Exercise 1.3

We start with the left-hand side (LHS) and show it is equal to the right-hand side (RHS).

$$\begin{aligned}
 \frac{f(x+h) - f(x)}{h} &= \frac{5^{x+h} - 5^x}{h} \\
 &= \frac{5^x \cdot 5^h - 5^x}{h} \\
 &= \frac{5^x(5^h - 1)}{h} \\
 &= 5^x \left(\frac{5^h - 1}{h} \right)
 \end{aligned}$$

The result is equal to the RHS, which completes the proof.

Exercise 1.4

(i) The composition table is completed according to the specified order for rows and columns (operation: row $\circ$ column).

- **Row Order:** $e, \tau, \tau', \tau'', \sigma', \sigma$

- **Column Order:** $e, \tau, \tau', \tau'', \sigma, \sigma'$

$\circ$	e	τ	τ'	τ''	σ	σ'
e	e	τ	τ'	τ''	σ	σ'
τ	τ	e	σ	σ'	τ'	τ''
τ'	τ'	σ'	e	σ	τ''	τ
τ''	τ''	σ	σ'	e	τ	τ'
σ'	σ'	τ'	τ''	τ	e	σ
σ	σ	τ''	τ	τ'	σ'	e

(ii) We find the order of each function (the smallest positive integer n such that $f^{\circ n} = e$):

- For e : $e^{\circ 1} = e \implies \mathbf{n = 1}$.
- For τ : $\tau^{\circ 2} = \tau \circ \tau = e \implies \mathbf{n = 2}$.
- For τ' : $\tau'^{\circ 2} = \tau' \circ \tau' = e \implies \mathbf{n = 2}$.
- For τ'' : $\tau''^{\circ 2} = \tau'' \circ \tau'' = e \implies \mathbf{n = 2}$.
- For σ : $\sigma^{\circ 2} = \sigma \circ \sigma = \sigma'$, and $\sigma^{\circ 3} = \sigma^{\circ 2} \circ \sigma = \sigma' \circ \sigma = e \implies \mathbf{n = 3}$.
- For σ' : $\sigma'^{\circ 2} = \sigma' \circ \sigma' = \sigma$, and $\sigma'^{\circ 3} = \sigma'^{\circ 2} \circ \sigma' = \sigma \circ \sigma' = e \implies \mathbf{n = 3}$.

Exercise 1.5

(i) To find the equation of the tangent line to the curve $y = \frac{1}{1+x^2}$ at the point $(-1, 1/2)$. First, we find the derivative:

$$y' = \frac{d}{dx}(1+x^2)^{-1} = -1(1+x^2)^{-2}(2x) = \frac{-2x}{(1+x^2)^2}$$

The slope m at $x = -1$ is:

$$m = \frac{-2(-1)}{(1+(-1)^2)^2} = \frac{2}{2^2} = \frac{1}{2}$$

Using the point-slope formula, the tangent line equation is:

$$y - \frac{1}{2} = \frac{1}{2}(x - (-1)) \implies \mathbf{y = \frac{1}{2}x + 1}$$

(ii) To find the equation of the tangent line to the curve $y = \frac{x}{1+x^2}$ at the point $(3, 0.3)$.
Using the quotient rule, the derivative is:

$$y' = \frac{(1)(1+x^2) - (x)(2x)}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

The slope m at $x = 3$ is:

$$m = \frac{1-3^2}{(1+3^2)^2} = \frac{-8}{10^2} = -\frac{8}{100} = -\frac{2}{25}$$

Using the point-slope formula, the tangent line equation is (since $0.3 = \frac{3}{10}$):

$$y - \frac{3}{10} = -\frac{2}{25}(x - 3) \implies y = -\frac{2}{25}x + \frac{6}{25} + \frac{3}{10} \implies y = -\frac{2}{25}x + \frac{27}{50}$$

Exercise 1.6

(a) For $y = x^2 f(x)$, using the Product Rule:

$$\frac{dy}{dx} = (x^2)'f(x) + x^2 f'(x) = 2x f(x) + x^2 f'(x)$$

(b) For $y = \frac{f(x)}{x^2}$, using the Quotient Rule:

$$\frac{dy}{dx} = \frac{f'(x) \cdot x^2 - f(x) \cdot 2x}{(x^2)^2} = \frac{x f'(x) - 2f(x)}{x^3}$$

(c) For $y = \frac{x^2}{f(x)}$, using the Quotient Rule:

$$\frac{dy}{dx} = \frac{2x f(x) - x^2 f'(x)}{[f(x)]^2}$$

(d) For $y = \frac{1+xf(x)}{\sqrt{x}}$, using the Quotient Rule:

$$\frac{dy}{dx} = \frac{(f(x) + x f'(x))\sqrt{x} - (1 + x f(x))(\frac{1}{2\sqrt{x}})}{x}$$

To simplify, multiply the numerator and denominator by $2\sqrt{x}$:

$$\frac{dy}{dx} = \frac{2x(f(x) + x f'(x)) - (1 + x f(x))}{2x\sqrt{x}} = \frac{x f(x) + 2x^2 f'(x) - 1}{2x^{3/2}}$$

Exercise 1.7

(i) To prove the Reciprocal Rule, we use the Quotient Rule with $u = 1$ and $v = g(x)$. This gives $u' = 0$ and $v' = g'(x)$.

$$\begin{aligned}\frac{d}{dx} \left[\frac{1}{g(x)} \right] &= \frac{(0) \cdot g(x) - (1) \cdot g'(x)}{[g(x)]^2} \\ &= -\frac{g'(x)}{[g(x)]^2}\end{aligned}$$

(ii) To verify the Power Rule for negative integers, let $y = x^{-n} = \frac{1}{x^n}$ for $n \in \mathbb{N}$. Using the Reciprocal Rule with $g(x) = x^n$, we have $g'(x) = nx^{n-1}$.

$$\begin{aligned}\frac{d}{dx}(x^{-n}) &= \frac{d}{dx} \left(\frac{1}{x^n} \right) \\ &= -\frac{g'(x)}{[g(x)]^2} \\ &= -\frac{nx^{n-1}}{(x^n)^2} \\ &= -\frac{nx^{n-1}}{x^{2n}} \\ &= -nx^{(n-1)-2n} \\ &= -nx^{-n-1}\end{aligned}$$